S.P.M College,Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1) **– Hira Kumar**

Web Development

HTML

Code/Text Editors

- Text editors are used to create and modify web pages.
- HTML, CSS, JavaScript etc. codes can be written in any text editors including the notepad. Some of the popular text editors are :-
- **For PC/Laptop (need install)**
  - ✓ Notepad
  - ✓ Notepad++ → https://github.com/notepad-plus-plus/notepad-plus plus/releases/download/v8.9.2/npp.8.9.2.Installer.x64.exe

- ✓ Subline text editor → https://www.sublimetext.com/download_thanks?target=win-x64
- ✓ VS Code etc.

- **For online Editors (no need to install)**
  - ✓ CodePen
  - ✓ JSFiddle
  - ✓ Replit
- **For android**
  - ✓ Acode – Code Editor
  - ✓ Dcoder, Compiler IDE
  - ✓ Spck Code Editor
  - ✓ Quoda Code Editor
  - ✓ Termux

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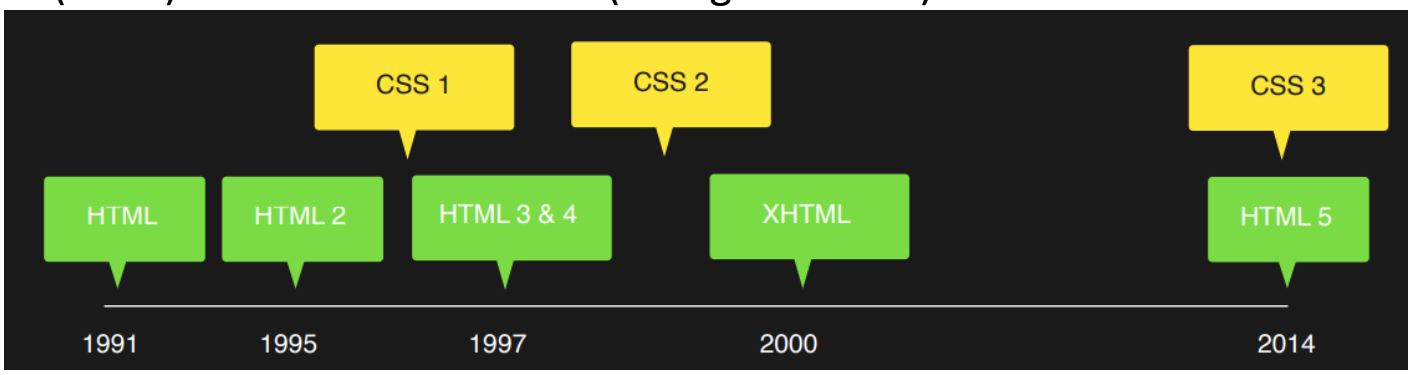
Detailed Course Contents:

- Introduction HTML
- HTML Basics
- HTML Elements
- HTML5 Semantic
- HTML Attributes
- HTML Headings
- HTML Paragraph
- HTML Styles
- HTML Formatting
- HTML Quotations
- HTML Computer Code
- HTML Comments & Colours
- HTML CSS, Links and Images
- HTML Lists
- HTML Blocks
- HTML Classes
- HTML Layout
- HTML Responsive
- HTML iframes
- HTML JavaScript
- HTML Head

❖ HTML

- HTML stands for HyperText Markup language.
- Hyper Text means text that contains links. It allows you to jump from one page to another page by clicking a link.These links are called hyperlinks. E.g. → <a href="**https://google.com**">Go to Google</a>
- A Markup Language is a language that uses tags to structure and format content. E.g. <h1> → Heading, <p> → Paragraph, <img> → Image, <a> → Link
- It's a markup language used to create web pages.
- This is the most basic building block of every web application, Without using HTML we cannot build web applications.

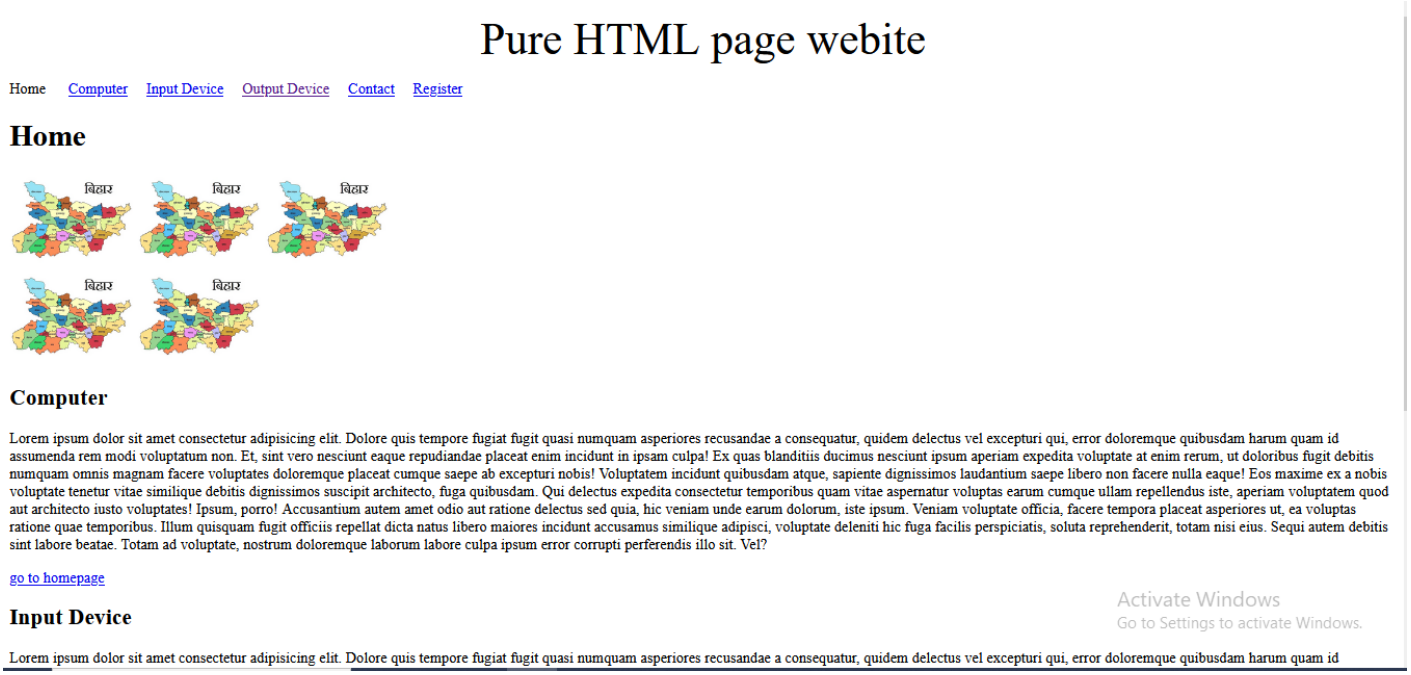
- First version of HTML is HTML in 1991 by Tim Berners-Lee, later it became known as HTML 1.0. Now the current standard used today (2026) is HTML5 from 2014. (Living Standard).



- File extension of html file is .html or .htm
- HTML is not a case sensitive.
- Example:

  <html>
  Or
  <HTML>
  Or
  <Html> all these are same effect.

Pure HTML page webite



❖ What are required software to create and run the html code.

1)Text Editor e.g. notepad, notepad++ etc.
2)Web browser e.g firefox, chrome IE etc...

❖ How to run a html code:

- ✓ Step1: Create a html file (extension of file should be .html or .htm)
- ✓ Step2: Right click on the html file and select open with notepad or any text editor.
- ✓ Step3: Write HTML code in the file and save it.
- ✓ Step4: To run html code right click on the html file and open with any web browser. Now you will see the output in the browser.

❖ Document Strcutre of HTML :-

In HTML, **document structure** refers to the organized layout of a web page, which defines how the content is arranged and displayed in the browser.

1. DOCTYPE Declaration (<!DOCTYPE html>)

- Tells the browser which version of HTML to use.
- Must be the very first line of your document.
- As of 2026, the current version of HTML is → HTML5
- For HTML5, it's simply declaration → <!DOCTYPE html>
- For HTML 1.0 → No doctype declaration used.
- For HTML2.0, it's long declaration → <!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0 //EN">
- For HTML4.01, it's long declaration → <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">

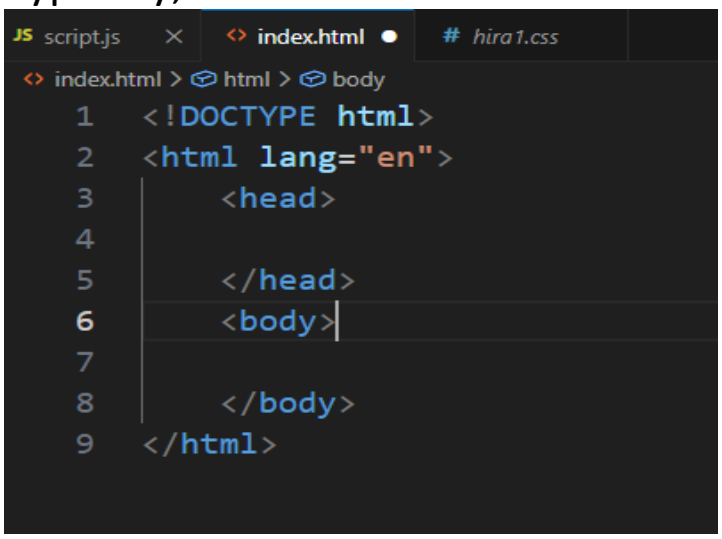
❖ Problem if you not use DOCTYPE declaration

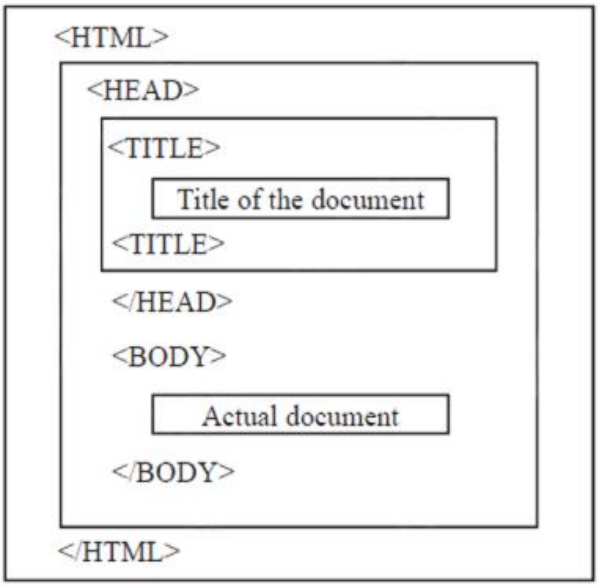
a. Layout issues
b. CSS not working properly
c. Inconsistent/Different design across browsers.

d. Box model problems
e. Old browser rendering behavior

2. HTML Element (<html>)

- The root element of the document.
- Wraps all content/element except the DOCTYPE
- Usually includes the lang attribute for accessibility.
- "It is not visible content; the browser does not display this section/element."
- Typically, it includes two child elements:- <head> and <body>





3. Head Section (<head>)

- The section is responsible for housing meta-information related to the document, encompassing/include details like the title, character encoding, and references to external resources.
- While these details are not visible on the web page itself, they hold significance (hold feature) for search engines and browsers.
- "It is not visible content; the browser does not display this section/element."
- Common elements that can be found in the section include:

✓ <title> → Page title (appears In browser tab - पेज का नाम browser में दिखाता हैं)
✓ <meta> → Character set, description, keyword (page का information देना)
✓ <link> → External CSS files connect only, not .js file connect (create by you or other source)

- ✓ <style> Write code for Internal CSS.
- ✓ <script> Write code for JavaScript.

```html
<!DOCTYPE html>
<html>
<head>
    <title>my college</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <link rel="stylesheet" href="hira1.css"> <!-- External Css file connect -
    <style>                                  <!-- Internal css file connect-->
        body
        {
            background-color: lightblue;
        }
    </style>
    <script> src="script.js"</script> <!--External JS file connect-->
    <script>
        console.log("Hello from Head Section"); <!-- internal js file connect
    </script>
        </script>
</head>
<body>

</body>
</html>
```

4. Body section (<body>)

- The <body> section contains the visible content of your web page, including text, images, forms, and interactive elements.
- "It is visible content; the browser display this section/element."
- This is where you structure the main content of your page using HTML elements like headings, paragraphs, lists, links, and more.
    - ✓ Heading (<h1> to <h6>)
    - ✓ Paragraphs (<p>)
    - ✓ Images (<img>)
    - ✓ Links (<a>)
    - ✓ Lists (<ul> <ol>)

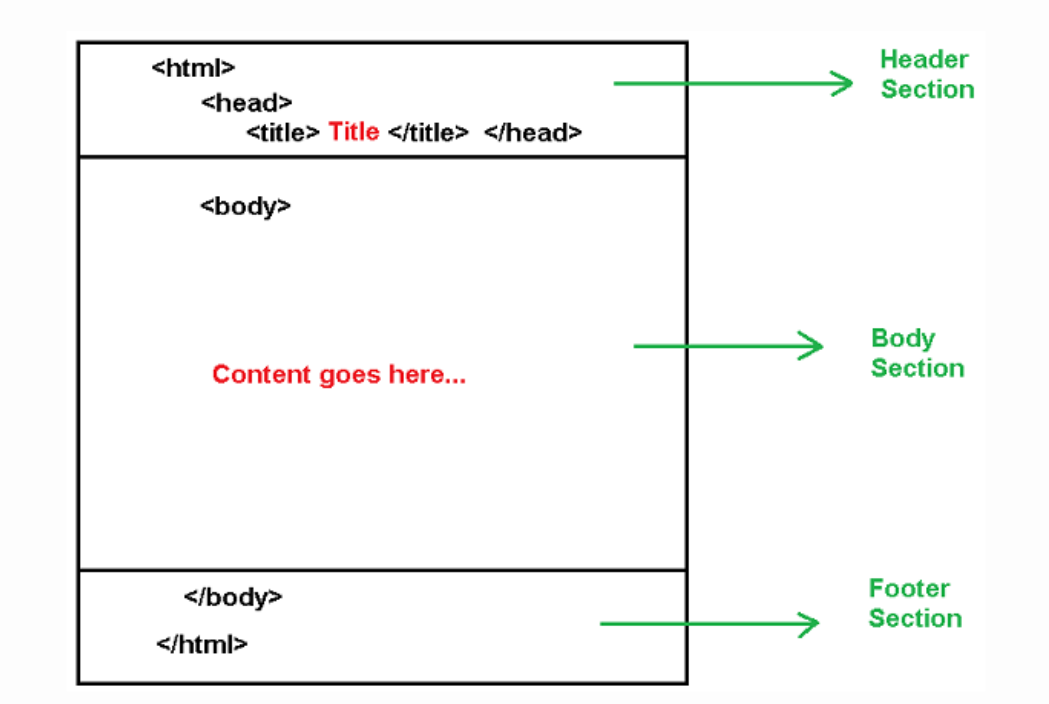
✓ Divisions ( )…..more

❖ HTML Web Page Semantic Structure

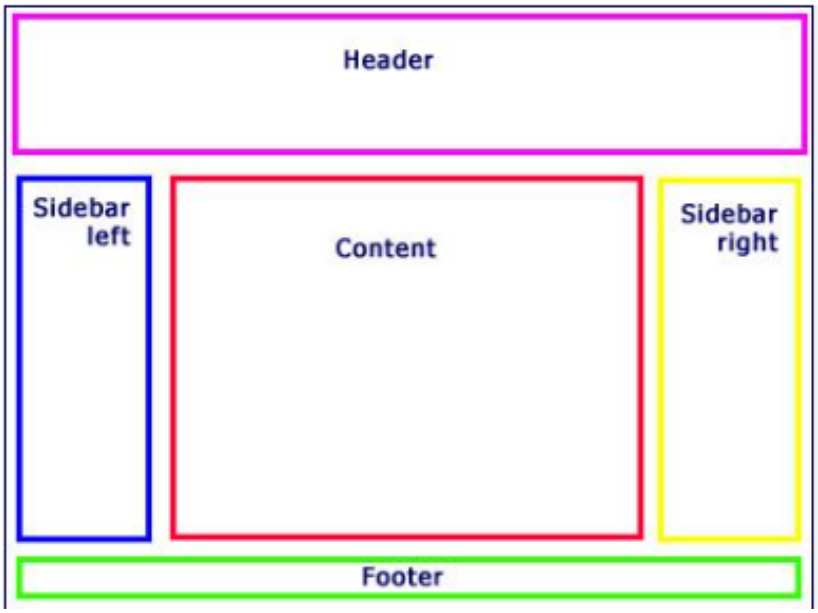
- An HTML page semantic structure is the basic layout or skeleton of a web page.
- It defines how content is organized inside a webpage.

❖ It is divided into 3 parts

1. **Header Section**
2. **Body Section / Main section**
3. **Footer section**



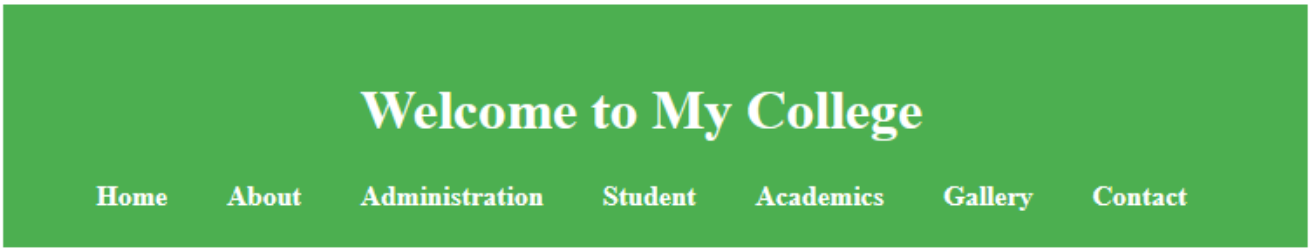
OR



1. Header section

- Usually placed at the top of the page
- Contains introductory content or navigation links – include → Logo, Company name, Website title, Navigation Menu, Search bar, Login/signup buttons, Social media links,…….

```
<!-- Header Section -->
<header>
    <h1>Welcome to My College</h1>
    <nav>
        <a href="#">Home</a>
        <a href="#">About</a>
        <a href="#">Administration</a>
        <a href="#">Student</a>
        <a href="#">Academics</a>
        <a href="#">Gallery</a>
        <a href="#">Contact</a>
    </nav>
</header>
```



➢ **Purpose**

✓ Shows website identity

✓ Helps user navigate

2. Body Section / Main Section

- Main content of the page, Visible to users, can include → Paragraphs <p>, Headings <h1>-<h6>, images <img>, tables <table>, articles <article>, sections <sections>,….
- Syntax → <main> content </main>

➢ **Purpose**

✓ Displays all the important content.

✓ User focuses on this part.

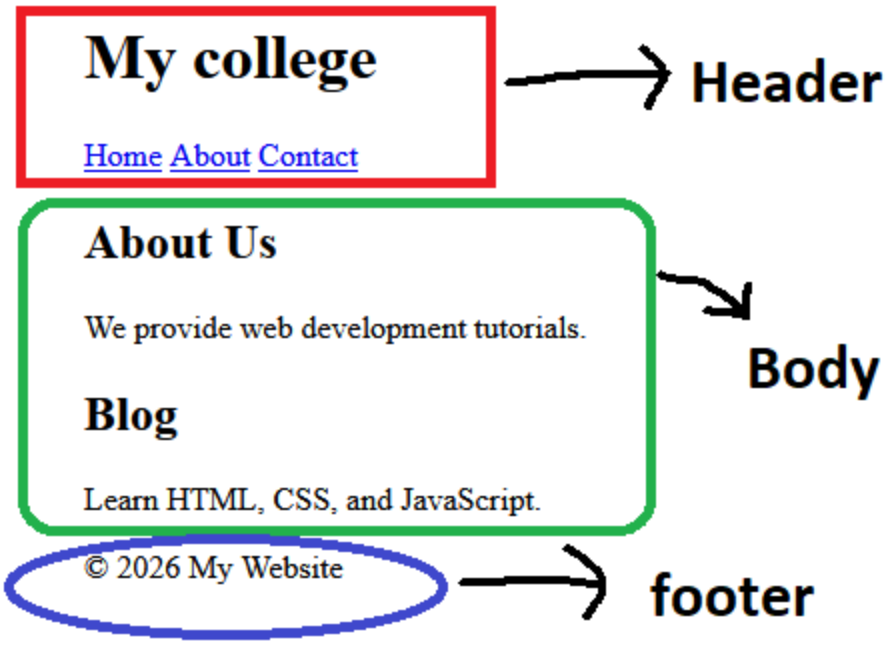
3. Footer Section (<footer>)

- Usually placed at the bottom of the page. Contains supplementary info. Can include → contact details, social media links, copyright info, Terms & privacy links,…..
- Syntax → <footer> content </footer>

➢ **Purpose**

✓ Provides extra information.

✓ Helps with website credits and navigation.



❖ Difference b/w HTML tags vs Elements

- HTML tags and HTML elements are related but distinct concepts in web development. Understanding the difference between them is essential for working with HTML effectively.

➢ HTML Tags:

Definition: HTML tags serve as the building blocks of an HTML document, delineating the start and end of an HTML element. These tags are enclosed within angle brackets, typically appearing in pairs which consist of an opening tag and a closing tag. The opening tag specifies the element's name, while the closing tag shares the same name but is preceded by a forward slash (/). For instance, the opening tag and closing tag are used together to define a paragraph element.

Purpose: Tags serve as markers that tell the browser how to interpret and render the content between them. They provide structure and formatting instructions to the web page.

➢ HTML Elements:

Definition: HTML elements are made up of an opening tag, closing tag, and the content (if any) that resides between them. An HTML element, as a whole, includes both the tags and the content enclosed within those tags. For example, Hello, World! is an HTML element that represents a paragraph containing the text "Hello, World!"

Purpose: HTML elements function as the fundamental constituents of an HTML document, playing a pivotal role in shaping and delivering the content of a web page. These elements encompass various aspects of a webpage's structure and content, encompassing essentials like headings, paragraphs, images, links, and a multitude of other components. Additionally, elements can be enriched by attributes that furnish supplementary information related to the element.

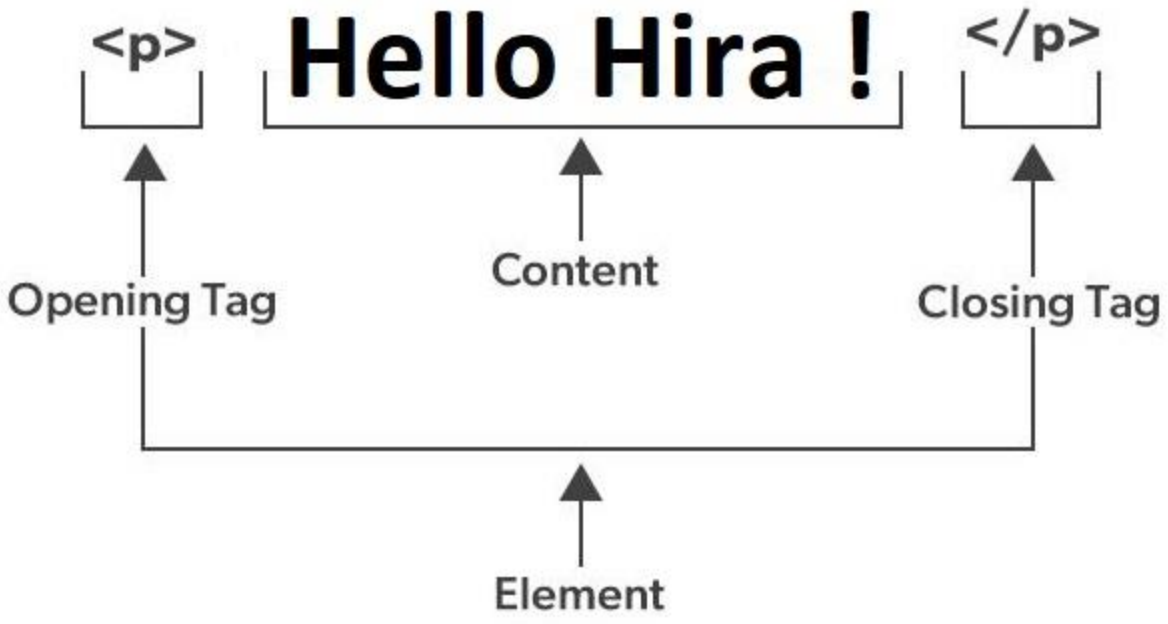
❖ Short Explanation

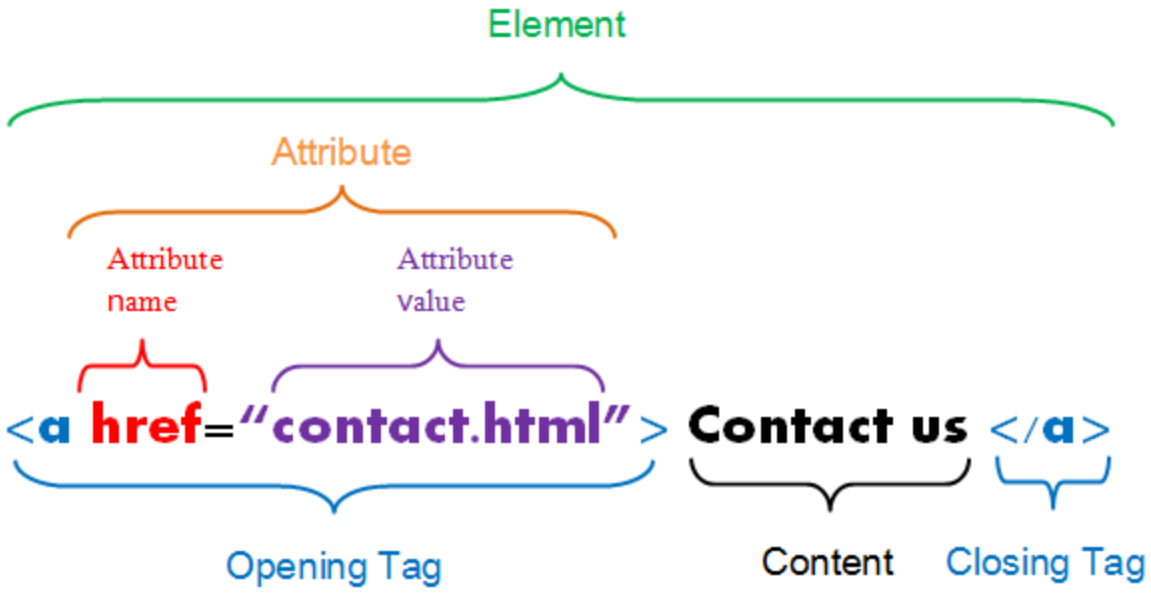
HTML Tag

- A tag is a piece of code that marks the start or end of an element. Written inside angle brackets < >.
- Two types of tags : a. Openin tag → starts an element e.g. →
  b. Closing tag → ends an element e.g. →
- Some tags are self-closing like → <img>, < br>,….

HTML Element

- An element is the complete structure that the browser interprets.
- E.g. → Hello World
  Here, → opening tag
  Hello word → content
   closing tage
  Entire thing hello word is the element.





❖ What is attribute ?

- Attribute is property that gives extra information about an HTML element and is written inside the opening tag.

OR

- Provide additional information about elements.

Example :- <video controls height="300" muted loop="infinite" poster="bihar.jpg">

e.g. - <img src="bike.jpg" alt="red bike" width="300" height ="200">

❖ Display property

- The display property defines how an HTML element appears on a web page. whether it takes full width (पूरा) (block) or only required (जितना जरूरत है) width (inline).

  In pure HTML (without CSS), elements are by default either:
- **Block-level elements**
- **Inline elements**

1. Block-Level Elements

- A block element Starts on a new line , Takes full width available, Can contain inline and other block elements.
- Each element appears on a new line.
- Example :-

<address>,<article>, <aside>,<blockquote>, <canvas>,<dd>, ,<dl>, <dt>,<fieldset>,<figcaption>,<figure>,<footer>,<form>,<h1> to <h6>, <header>,<hr>,<li>,<main>,<nav>,<noscript>,<ol>,<p>,<pre>,<section>, <table>,<tfoot>,<ul>,<video>

2. Inline Elements

- An inline element does not start on a new line, Takes only required width and Cannot contain block elements.
- All appear in the same line.
- Example :-

<a>,<abbr>,<b>,<bdo>,
,<button>,<cite>,<code>,<dfn>,,<i>,<i mg>,<input>,<label>,<map>,<mark>,<object>,<output>,<q>,<samp>,<s

cript>,<select>,<small>,,,<sub>,<sup>,<textarea>,<time>,<tt>,<var>

❖ Types of HTML tag

HTML tags are mainly two types.

1. **Paired Tag (container tag)**
2. **Unpaired Tag (Empty tag)**

1. Paired Tag

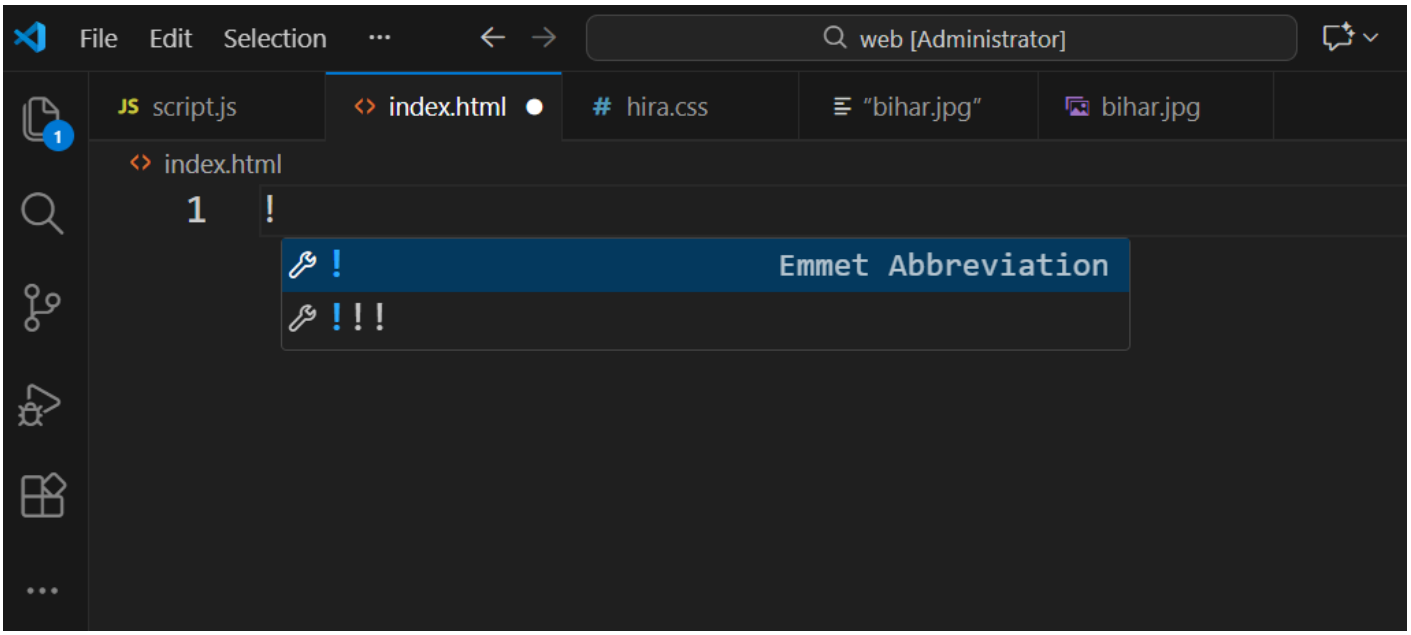
- A paired tag has both opening and closing tags.
- Syntax → <tagname> content </tagname>
- Example → content

2. Unpaired Tag

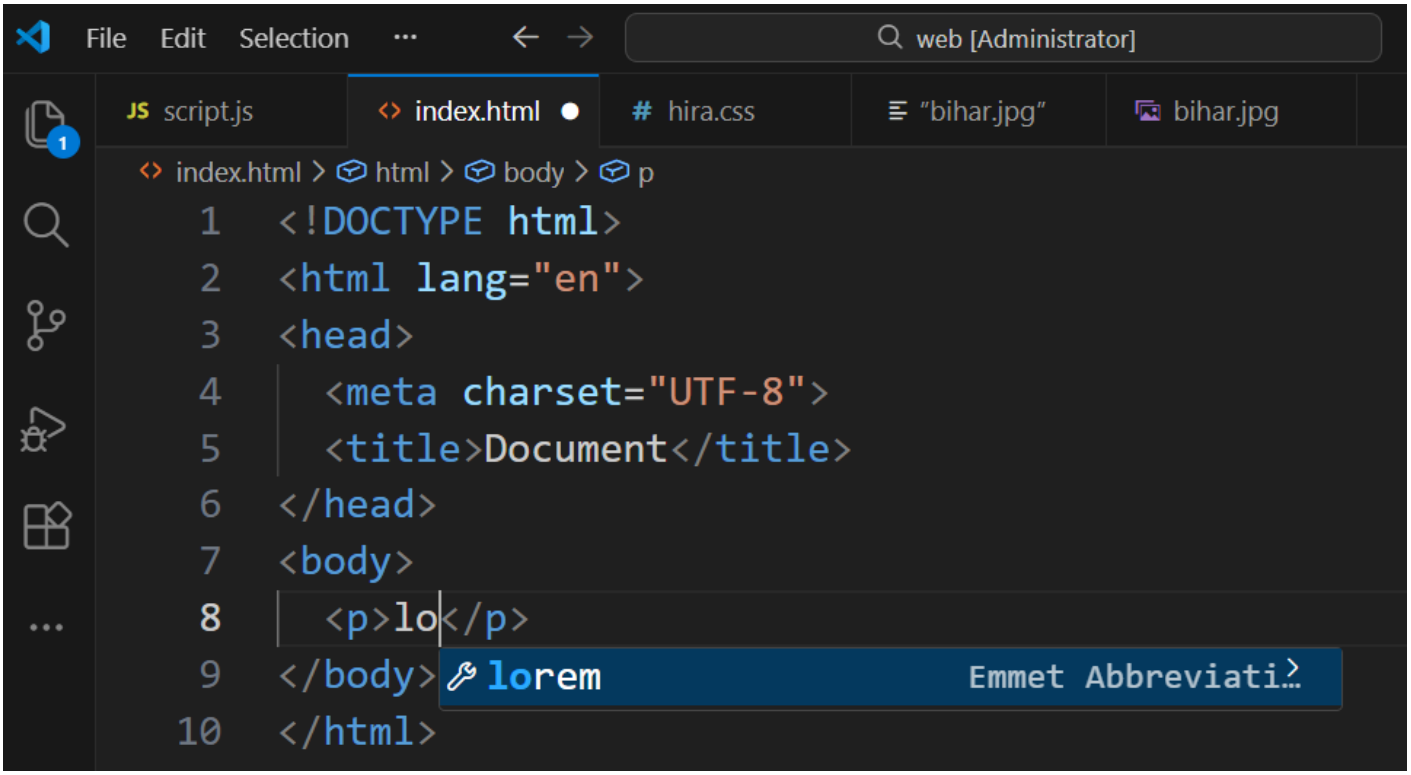
- An unpaired tag has no closing tag.
- Example →
, <image>, <hr>, <input>,…

Key Points to Remember

1. **Nesting**
   Elements must be properly nested (closed in reverse order)
2. **Self-closing tags**
   Elements like <img>, <br>, <hr> don't need closing tags.
3. **Comments**
   Use <!--comment write --> for notes
4. **VS Code users**
   Use **emmt** for all essential section for require in html (e.g. html,head,body>
   Use → press ! and click emmt



And use “loren” for random sentence.



अच्छी आदतें शुरू से ही डालें - सही तरीका सीखेंगे तो आगे बड़े project में आसानी होगी

HIRA KUMAR